

Pinned group summary: SO-n4-q1

Pinning-Sympy automated output

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1 Basic group attributes

Group	SO-n4-q1
Matrix size	4×4
Rank	1
Defining equations	$X^T B X = B$ and $\det(X) = 1$

1.1 Symmetric bilinear form

Dimension	4
Witt index	1
Primitive element	N/A
Epsilon	N/A

Matrix:
$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & c_0 & 0 \\ 0 & 0 & 0 & c_1 \end{bmatrix}$$

1.2 Lie algebra

The Lie algebra is denoted $\mathfrak{g}$. The defining equations for the $\mathfrak{g}$ are: $X^T B + B X = 0$ and $\text{tr}(X) = 0$
A generic element of $\mathfrak{g}$ is:

$$\begin{bmatrix} x_{00} & 0 & -c_0 x_{21} & -c_1 x_{31} \\ 0 & -x_{00} & -c_0 x_{20} & -c_1 x_{30} \\ x_{20} & x_{21} & 0 & x_{23} \\ x_{30} & x_{31} & -\frac{c_0 x_{23}}{c_1} & 0 \end{bmatrix}$$

1.3 Torus

The torus is denoted T . It has rank 1. A generic element of the T is

$$\begin{bmatrix} t_0 & 0 & 0 & 0 \\ 0 & \frac{1}{t_0} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

A **character** of T is a group homomorphism from T to the multiplicative group of the ground field. The set of characters is denoted $X_*(T)$. In this document, elements of $X_*(T)$ are written as row vectors $\alpha = (a_1, a_2, \dots, a_n)$ and such a vector is viewed as a function on diagonal matrices in the following way: for $\alpha = (a_1, a_2, \dots, a_n) \in X_*(T)$ and $t = \text{diag}(t_1, t_2, \dots, t_n) \in T$,

$$\alpha(t) = \prod_{i=1}^n t_i^{a_i} = t_1^{a_1} t_2^{a_2} \dots t_n^{a_n}$$

The rows of the matrix below are trivial characters, i.e. elements $\alpha \in X_*(T)$ which are the identity function on T . It would be more accurate to describe $X_*(T)$ as a quotient of a set of group homomorphisms modulo the lattice generated by trivial characters, but this distinction is not critical for our purposes.

$$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

2 Root system

Dynkin type	A_1
Irreducible	True
Reduced	True
Simply laced	True
Number of roots	2

2.1 Dynkin diagram

The root system is irreducible and has Dynkin diagram:

○

2.2 Cartan matrix

The root system is irreducible and has Cartan matrix:

[2]

2.3 Roots

Definition 2.1. A root $\alpha \in \Phi$ is **multipliable** if $2\alpha \in \Phi$.

Root	Short Form	Norm ²	Simple	Sign	Multipliable
(1, 0, 0, 0)	(1,)	1	Yes	+	No
(-1, 0, 0, 0)	(-1,)	1	No	-	No

2.4 Coroots

Given a root system Φ , for each root $\alpha \in \Phi$, we find a coroot $\alpha^\vee$ (an integer vector of the same length) so that $\langle \alpha, \alpha^\vee \rangle = 2$. There is not necessarily a unique choice of $\alpha^\vee$, but for the purposes of this document, we choose $\alpha^\vee$ so that it is the integer vector of minimal Euclidean norm among those satisfying $\langle \alpha, \alpha^\vee \rangle = 2$.

Root	Coroot
(-1, 0, 0, 0)	(-2, 2, 0, 0)
(1, 0, 0, 0)	(2, -2, 0, 0)

3 Root spaces

3.1 Dimensions

For a root $\alpha \in \Phi$, the dimension of the associated root space V_α (and also the dimension of the root subgroup U_α) is denoted d_α . The table below lists these dimensions.

α	d_α
$(-1, 0, 0, 0)$	2
$(1, 0, 0, 0)$	2

3.2 Root space table

The table below lists each root along with a generic element of the associated root space (inside the Lie algebra). These are computed as follows: given a character α , consider the set

$$V_\alpha = \{X \in \mathfrak{g} : t \cdot X \cdot t^{-1} = \alpha(t) \cdot X, \forall t \in T\}$$

This is a vector subspace of the Lie algebra $\mathfrak{g}$. When V_α is more than just the zero vector, α is called a **root** and the root system Φ is the set of all such roots. For this document, roots are computed as follows: compile a list of potential character vectors α , which are all possible integer vectors of length n with entries from $\{0, \pm 1, \pm 2\}$, then reduce that list of potentials by quotienting by the trivial characters of the torus. Next, instantiate a generic torus element t and a generic Lie algebra element X . For each potential character α , solve the equation

$$t \cdot X \cdot t^{-1} = \alpha(t) \cdot X$$

for X . If there is a nonzero solution space, add α to the set of roots.

Root	Generic element of root space
$(-1, 0, 0, 0)$	$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -c_0 u_1 & -c_1 u_0 \\ u_1 & 0 & 0 & 0 \\ u_0 & 0 & 0 & 0 \end{bmatrix}$
$(1, 0, 0, 0)$	$\begin{bmatrix} 0 & 0 & -c_0 u_0 & -c_1 u_1 \\ 0 & 0 & 0 & 0 \\ 0 & u_0 & 0 & 0 \\ 0 & u_1 & 0 & 0 \end{bmatrix}$

4 Root subgroups

4.1 Root subgroup table

For the purposes of this document, root subgroups are found by simply exponentiating the associated root space. That is, for a root $\alpha \in \Phi$, the root space is

$$V_\alpha = \{X \in \mathfrak{g} : t \cdot X \cdot t^{-1} = \alpha(t) \cdot X, \forall t \in T\}$$

and the root subgroup is

$$U_\alpha = \left\{ e^X = 1 + X + \frac{X^2}{2!} + \frac{X^3}{3!} + \cdots : X \in V_\alpha \right\}$$

For the purposes of the groups considered in this package, the X^3 and higher terms of the exponential series always vanish, and even the quadratic term frequently vanishes, so we can simplify the description of U_α to:

$$U_\alpha = \left\{ 1 + X + \frac{X^2}{2} : X \in V_\alpha \right\}$$

The table below lists each root along with a generic element of the associated root subgroup.

Root	Generic element of root subgroup
$(-1, 0, 0, 0)$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ -\frac{c_0 u_1^2}{2} - \frac{c_1 u_0^2}{2} & 1 & -c_0 u_1 & -c_1 u_0 \\ u_1 & 0 & 1 & 0 \\ u_0 & 0 & 0 & 1 \end{bmatrix}$
$(1, 0, 0, 0)$	$\begin{bmatrix} 1 & -\frac{c_0 u_0^2}{2} - \frac{c_1 u_1^2}{2} & -c_0 u_0 & -c_1 u_1 \\ 0 & 1 & 0 & 0 \\ 0 & u_0 & 1 & 0 \\ 0 & u_1 & 0 & 1 \end{bmatrix}$

4.2 Homomorphism property

For each root $\alpha \in \Phi$, the associated root subgroup map $X_\alpha : V_\alpha \rightarrow G$ is a pseudo-homomorphism with respect to the additive structure on V_α and the group operation (i.e. matrix multiplication) in G . That is, for any $u, v \in V_\alpha$,

$$X_\alpha(u) \cdot X_\alpha(v) = X_\alpha(u + v) \cdot \prod_{\substack{i > 1 \\ i\alpha \in \Phi}} X_{i\alpha} \left(q_\alpha^i(u, v) \right) \quad (1)$$

for some homogeneous polynomial maps $q_\alpha^i : V_\alpha \times V_\alpha \rightarrow V_{i\alpha}$. Thus X_α is a group homomorphism if and only the product on the right hand side reduces to 1. The quantities $q_\alpha^i(u, v)$ are called **homomorphism defect coefficients**. For any root α of a root system Φ of any group considered in the Pinning-Sympy package, the set

$$\{i\alpha \in \Phi : i \in \mathbb{Z}_{\geq 2}\}$$

is either empty or a singleton set consisting of $\{2\alpha\}$. The non-empty case only arises when Φ is a non-reduced root system (of Dynkin type BC), and when α is a multipliable root.

Since the root system for this group is reduced, there are no multipliable roots and the product is always empty, hence each X_α is a group homomorphism there are no homomorphism defect coefficients.

5 Commutators

The commutator formula says that for two roots $\alpha, \beta \in \Phi$, the commutator of two root group elements $X_\alpha(u)$ and $X_\beta(v)$ is controlled by the structure of Φ , in particular by which positive integer linear combinations $i\alpha + j\beta$ are elements of Φ . Recall that

$$(\alpha, \beta) = \{i\alpha + j\beta \in \Phi : i, j \in \mathbb{Z}_{\geq 1}\}$$

Specifically, we have the commutator formula

$$\left[X_\alpha(u), X_\beta(v) \right] = \prod_{(\alpha, \beta)} X_{i\alpha + j\beta} \left(N_{ij}^{\alpha\beta}(u, v) \right)$$

for some homogeneous polynomial functions $N_{ij}^{\alpha\beta} : V_\alpha \times V_\beta \rightarrow V_{i\alpha + j\beta}$. The root system for this particular group has no pairs of roots α, β where the set (α, β) is non-empty, so the commutator of any two root subgroup elements $X_\alpha(u)$ and $X_\beta(v)$ is trivial, i.e. all root subgroups in this group commute with each other.

6 Weyl group

One of the defining characteristics of a root system Φ is that for any two roots $\alpha, \beta \in \Phi$, the reflection of β across the hyperplane perpendicular to α is another root. The reflection across the hyperplane perpendicular to α is denoted σ_α , and the previous sentence can be rephrased as saying that σ_α is a (bijective) function from Φ to itself. The reflection of β is then denoted $\sigma_\alpha(\beta)$. For each root $\alpha \in \Phi$, there is an element $w_\alpha \in G$ with the following properties:

1. $w_\alpha \in N_G(T)$, i.e. w_α normalizes the torus, i.e. $w_\alpha \cdot t \cdot w_\alpha^{-1} \in T$ for all $t \in T$.
2. $w_\alpha^2 \in Z_G(T)$, i.e. w_α^2 centralizes the torus, i.e. $w_\alpha^2 \cdot t \cdot (w_\alpha^2)^{-1} = t$ for all $t \in T$. Under some assumptions, $T = Z_G(T)$, though this is not true for all of the groups considered in this package.
3. For any root $\beta \in \Phi$, the conjugation of the root subgroup U_β by w_α is the root subgroup $U_{\sigma_\alpha(\beta)}$. That is, for $x \in U_\beta$,

$$w_\alpha \cdot x \cdot w_\alpha^{-1} \in U_{\sigma_\alpha(\beta)}$$

More precisely, given an element $x_\beta(u) \in U_\beta$, there is some function $\varphi_{\alpha\beta}(u)$ so that

$$w_\alpha \cdot x_\beta(u) \cdot w_\alpha^{-1} = x_{\sigma_\alpha(\beta)}(\varphi_{\alpha\beta}(u))$$

Taken together, properties (1) and (2) above say that viewed as an element of the quotient group $N_G(T)/Z_G(T)$, (the class of) w_α is an element of order 2 (or order 1 but that doesn't happen for other reasons).

6.1 Weyl group elements

The table below lists roots along with an associated Weyl group element.

α	w_α
$(-1,)$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$
$(1,)$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$

6.2 Weyl group conjugation coefficients

The table below lists roots along with the coefficients obtain in performing conjugation by an associated Weyl element.

α	β	$\gamma = \sigma_\alpha(\beta)$	$\varphi_{\alpha\beta}(u)$
$(-1,)$	$(-1,)$	$(1,)$	$\begin{bmatrix} u_1 & -u_0 \end{bmatrix}$
$(-1,)$	$(1,)$	$(-1,)$	$\begin{bmatrix} -u_1 & u_0 \end{bmatrix}$
$(1,)$	$(-1,)$	$(1,)$	$\begin{bmatrix} u_1 & -u_0 \end{bmatrix}$
$(1,)$	$(1,)$	$(-1,)$	$\begin{bmatrix} -u_1 & u_0 \end{bmatrix}$

6.3 Weyl group conjugation identities

$$\alpha = (-1, 0, 0, 0), \beta = (-1, 0, 0, 0), \sigma_\alpha(\beta) = (1, 0, 0, 0)$$

$$w_\alpha X_\beta \left(\begin{bmatrix} u_0 \\ u_1 \end{bmatrix} \right) w_\alpha^{-1} = X_{\sigma_\alpha(\beta)} \left(\begin{bmatrix} u_1 \\ -u_0 \end{bmatrix} \right)$$

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -\frac{c_0 u_1^2}{2} - \frac{c_1 u_0^2}{2} & 1 & -c_0 u_1 & -c_1 u_0 \\ u_1 & 0 & 1 & 0 \\ u_0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & -\frac{c_0 u_1^2}{2} - \frac{c_1 u_0^2}{2} & -c_0 u_1 & c_1 u_0 \\ 0 & 1 & 0 & 0 \\ 0 & u_1 & 1 & 0 \\ 0 & -u_0 & 0 & 1 \end{bmatrix}$$

$$\alpha = (-1, 0, 0, 0), \beta = (1, 0, 0, 0), \sigma_\alpha(\beta) = (-1, 0, 0, 0)$$

$$w_\alpha X_\beta \left(\begin{bmatrix} u_0 \\ u_1 \end{bmatrix} \right) w_\alpha^{-1} = X_{\sigma_\alpha(\beta)} \left(\begin{bmatrix} -u_1 \\ u_0 \end{bmatrix} \right)$$

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & -\frac{c_0 u_0^2}{2} - \frac{c_1 u_1^2}{2} & -c_0 u_0 & -c_1 u_1 \\ 0 & 1 & 0 & 0 \\ 0 & u_0 & 1 & 0 \\ 0 & u_1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -\frac{c_0 u_0^2}{2} - \frac{c_1 u_1^2}{2} & 1 & -c_0 u_0 & c_1 u_1 \\ u_0 & 0 & 1 & 0 \\ -u_1 & 0 & 0 & 1 \end{bmatrix}$$